Computational Biology Assignment 2

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Viterbi algorithm

Emission probabilities (B)

	A	U	G	C
E	0.25	0.25	0.25	0.25
I	0.4	0.4	0.05	0.15

(1)

Transition probabilities (A)

	E	I
E	0.9	0.1
I	0.2	0.8

(2)

Initial probabilities (Π)

$$\Pi(s) = (\Pi(E) = 0.5,\ \Pi(I) = 0.5) \tag{3}$$

Initialisation ($t = 1$)

$\delta_t(s)$ - max probability of any path ending in that state.
$\psi_t(s)$ - The previous state that gave the max probability
$\delta_1(s) = \Pi(s) * B(O_1|s)$

Recursion ($t \geq 2$)

$$\delta_t(s) = \underset{r \in s}{max}(\delta_{t-1}(r) * A(r|s) * B(O_i|s)) \tag{4}$$

$$\psi_t(s) = \underset{r \in s}{argmax}(\delta_t(s)) \tag{5}$$

Termination (backtracking)

p_t - Probability of most probable s at t
s_t^* - Most probable state for t

$$p_t = max_s(\delta_t(s)) \tag{6}$$
$$s_T^* = argmax_s(\delta_T(s)) \tag{7}$$
$$s_t^* = \psi_{t+1}(s_{t+1}^*) \quad for \quad t = T-1 \; down \; to \; 1 \tag{8}$$

Model Definitions and Notations

We use Hill-type regulatory functions for activation and repression:

$$h_+(p;\theta,n) = \frac{p^n}{\theta^n + p^n}, \tag{9}$$
$$h_-(p;\theta,n) = \frac{\theta^n}{\theta^n + p^n} \; = \; 1 - h_+(p;\theta,n). \tag{10}$$

Here, h_+ is an activation Hill function and h_- is a repression Hill function.

Parameter meanings.

- m_A, m_B: maximal regulated transcription rates.
- a_A, a_B: basal transcription rates (used in Mechanism II / SDE formulation).
- γ_A, γ_B: degradation rates of spliced mRNA (or mRNA in Mechanism I).
- β_A, β_B: baseline splicing rates.
- k_{PA}, k_{PB}: translation rates from spliced mRNA to protein.
- δ_{PA}, δ_{PB}: protein degradation rates.
- θ_A, θ_B: Hill thresholds (half-activation / half-repression concentrations).
- n_A, n_B: Hill coefficients controlling nonlinearity/steepness.
- $\sigma_{1A}, \sigma_{2A}, \sigma_{1B}, \sigma_{2B}$: noise intensities in the SDE model.

Noise notation (SDE model only). $B_{1A}(t), B_{2A}(t), B_{1B}(t), B_{2B}(t)$ denote (independent) standard Brownian motions (Wiener processes).

Mechanism I: Transcriptional regulation (4D ODE)

Normal Regulation

In the normal case, Protein B activates transcription of Gene A, and Protein A represses transcription of Gene B:

$$\frac{dr_A}{dt} = m_A\, h_+(p_B; \theta_B, n_B) - \gamma_A\, r_A, \tag{11}$$
$$\frac{dr_B}{dt} = m_B\, h_-(p_A; \theta_A, n_A) - \gamma_B\, r_B, \tag{12}$$
$$\frac{dp_A}{dt} = k_{PA}\, r_A - \delta_{PA}\, p_A, \tag{13}$$
$$\frac{dp_B}{dt} = k_{PB}\, r_B - \delta_{PB}\, p_B. \tag{14}$$

Disabled Repression of Gene B

In the disabled case of Mechanism I, the transcriptional repression of Gene B by Protein A is disrupted. Therefore, the transcription rate of r_B no longer depends on p_A:

$$\frac{dr_A}{dt} = m_A\, h_+(p_B; \theta_B, n_B) - \gamma_A\, r_A, \tag{15}$$
$$\frac{dr_B}{dt} = m_B - \gamma_B\, r_B, \tag{16}$$
$$\frac{dp_A}{dt} = k_{PA}\, r_A - \delta_{PA}\, p_A, \tag{17}$$
$$\frac{dp_B}{dt} = k_{PB}\, r_B - \delta_{PB}\, p_B. \tag{18}$$

Mechanism II: Splicing sabotage (6D)

Mechanism II models disruption at the *splicing stage*, where Protein A modulates the maturation of Gene B pre-mRNA into functional mRNA.

Regulated rates (shared by ODE/SDE formulations)

We define the time-varying transcription and effective splicing rates as

$$\alpha_A(t) = a_A + m_A\, h_+(p_B; \theta_B, n_B), \tag{19}$$
$$\alpha_B(t) = a_B + m_B\, h_-(p_A; \theta_A, n_A), \tag{20}$$

$$\beta_A^*(t) = \beta_A\, h_+(p_B; \theta_B, n_B), \tag{21}$$
$$\beta_B^*(t) = \beta_B\, h_-(p_A; \theta_A, n_A). \tag{22}$$

Deterministic ODE formulation (drift only)

The deterministic part of Mechanism II is:

$$\frac{dU_A}{dt} = \alpha_A(t) - \beta_A^*(t)\, U_A, \tag{23}$$

$$\frac{dS_A}{dt} = \beta_A^*(t)\, U_A - \gamma_A\, S_A, \tag{24}$$

$$\frac{dp_A}{dt} = k_{PA}\, S_A - \delta_{PA}\, p_A, \tag{25}$$

$$\frac{dU_B}{dt} = \alpha_B(t) - \beta_B^*(t)\, U_B, \tag{26}$$

$$\frac{dS_B}{dt} = \beta_B^*(t)\, U_B - \gamma_B\, S_B, \tag{27}$$

$$\frac{dp_B}{dt} = k_{PB}\, S_B - \delta_{PB}\, p_B. \tag{28}$$

Stochastic SDE formulation (drift + noise)

The SDE model adds intrinsic noise to the unspliced and spliced mRNA dynamics:

$$dU_A = \big(\alpha_A(t) - \beta_A^*(t)\, U_A\big)\, dt + \sigma_{1A}\, dB_{1A}(t), \tag{29}$$

$$dS_A = \big(\beta_A^*(t)\, U_A - \gamma_A\, S_A\big)\, dt + \sigma_{2A}\, dB_{2A}(t), \tag{30}$$

$$dU_B = \big(\alpha_B(t) - \beta_B^*(t)\, U_B\big)\, dt + \sigma_{1B}\, dB_{1B}(t), \tag{31}$$

$$dS_B = \big(\beta_B^*(t)\, U_B - \gamma_B\, S_B\big)\, dt + \sigma_{2B}\, dB_{2B}(t). \tag{32}$$

Protein dynamics are kept deterministic:

$$\frac{dp_A}{dt} = k_{PA}\, S_A - \delta_{PA}\, p_A, \tag{33}$$

$$\frac{dp_B}{dt} = k_{PB}\, S_B - \delta_{PB}\, p_B. \tag{34}$$

Remark. The deterministic ODE system for Mechanism II is exactly the drift part of the SDE system (i.e., obtained by setting all noise intensities $\sigma_{1A}, \sigma_{2A}, \sigma_{1B}, \sigma_{2B}$ to zero).

Bonus: Modeling Downstream Metabolic Effects

(a) Biological interpretation of each term

$$\frac{dR}{dt} = \alpha R - \beta RE \tag{35}$$

- αR: net growth of the cellular resource.
  - If $E = 0$, then R grows exponentially at rate α.
- $-\beta RE$: depletion of resource due to enzyme-mediated consumption.
  - The product RE means consumption is stronger when both resource and enzyme are abundant.
  - β controls how efficiently E consumes R.

$$\frac{dE}{dt} = -\gamma E + \delta RE. \tag{36}$$

- $-\gamma E$: natural degradation of the enzyme.
  - If $R = 0$, then E decays exponentially at rate γ.
- $+\delta RE$: enzyme accumulation supported by the available resource.
  - More resource R enables stronger downstream metabolic activity, increasing E.
  - δ controls how strongly resource availability boosts enzyme activity.

(b) Stability analysis

Based on the given system, we obtain the equilibrium points as follows:

$$\frac{dR}{dt} = R(\alpha - \beta E) \tag{37}$$

$$\frac{dE}{dt} = E(-\gamma + \delta R) \tag{38}$$

$$R^*(\alpha - \beta E^*) = 0, \tag{39}$$

$$E^*(-\gamma + \delta R^*) = 0 \tag{40}$$

From above, either $R^* = 0$ or $E^* = \alpha/\beta$, and either $E^* = 0$ or $R^* = \gamma/\delta$.

Thus: The trivial equilibrium is $(R^*, E^*) = (0, 0)$, and the coexistence equilibrium is: $(R^*, E^*) = (\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$

Substituting the given parameter values into the second equilibrium point, we get:

$$(R^*, E^*) = (\frac{1}{0.9}, \frac{2}{1.1}) \tag{41}$$

Jacobian Matrix

The Jacobian for this system is defined as:

$$J(R,E) = \begin{pmatrix} \alpha - \beta E & -\beta R \\ \delta E & -\gamma + \delta R \end{pmatrix} \tag{42}$$

Evaluating it at the trivial equilibrium point results in a matrix:

$$J(0,0) = \begin{pmatrix} \alpha & 0 \\ 0 & -\gamma \end{pmatrix} \tag{43}$$

From which we can see that the eigenvalues are: $\lambda_1 = \alpha$, $\lambda_2 = -\gamma$. With $\alpha = 2$ and $\gamma = 1$, we obtain $\lambda_1 = 2 > 0$ and $\lambda_2 = -1 < 0$, so $(0,0)$ is a saddle point and thus unstable.

For the Jacobian of the second equilibrium point, we can first substitute the values to then obtain a simplified form as such:

$$J(R^*,E^*) = \begin{pmatrix} 0 & -\beta R^* \\ \delta E^* & 0 \end{pmatrix} \tag{44}$$

We can then calculate the eigenvalues from the determinant:

$$\lambda^2 = -\beta\delta R^* E^* = -\beta\delta \left(\frac{\gamma}{\delta}\right)\left(\frac{\alpha}{\beta}\right) = -\alpha\gamma \tag{45}$$

,

$$\lambda = \pm i\sqrt{\alpha\gamma} \tag{46}$$

For $\alpha = 2$ and $\gamma = 1$, this gives

$$\lambda = \pm i\sqrt{2}, \tag{47}$$

thus indicating stable oscillations.